

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5

7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sandra Sudevan

RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
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Experiment 1: Conversion to 1NF

AIM:

To convert an unnormalized relation into **First Normal Form (1NF)**.

PROCEDURE:

1. Identify repeating groups and multivalued attributes.
2. Separate repeating groups into individual rows.
3. Ensure every attribute contains only atomic values.
4. Identify the primary key.
5. Verify that there are no repeating groups.

Given:

STUDENT(SID, Name, Phone, Course)

If a student has multiple phone numbers/courses, the values are repeated.

Before 1NF:

101 | Arun | 9876,8765 | DBMS,OS

102 | Ravi | 7654 | CN

Convert multivalued attributes into atomic values.

After 1NF:

101 | Arun | 9876 | DBMS

101 | Arun | 8765 | OS

102 | Ravi | 7654 | CN

RESULT:

The unnormalized relation was successfully converted into **1NF**.

Experiment 2: Decomposition to 2NF

AIM:

To analyze the given relation and eliminate partial dependencies to achieve **2NF**.

PROCEDURE:

1. List the given functional dependencies.
2. Find the candidate key using attribute closure.
3. Identify partial dependencies.
4. Separate attributes depending on part of the composite key.
5. Decompose the relation into smaller relations.
6. Verify that all relations satisfy 2NF.

Given:

$R(A,B,C,D,E)$

FDs:

$$A \rightarrow BC$$

$$BC \rightarrow D$$

$$D \rightarrow E$$

Find closure:

$$A^+ = \{A\}$$

$$A \rightarrow BC$$

$$A^+ = \{A,B,C\}$$

$$BC \rightarrow D$$

$$A^+ = \{A,B,C,D\}$$

$$D \rightarrow E$$

$$A^+ = \{A,B,C,D,E\}$$

Therefore:

Candidate Key = A

RESULT:

The relation was successfully analyzed and decomposed into **2NF**, eliminating partial dependencies.

Experiment 3: Decomposition to 3NF**AIM:**

To normalize the EMPLOYEE relation into **Third Normal Form (3NF)** by removing transitive dependencies.

PROCEDURE:

1. Identify the functional dependencies.
2. Determine the candidate key.
3. Identify transitive dependencies.
4. Separate the attributes causing transitive dependency.
5. Create separate relations with appropriate primary keys.
6. Verify that the resulting relations satisfy 3NF.

Given:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

FDs:

$\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Transitive dependency:

$\text{EmpID} \rightarrow \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}, \text{ManagerID}$

Decompose:

EMPLOYEE(EmpID, EmpName, DeptID)

DEPARTMENT(DeptID, DeptName, ManagerID)

Thus, the transitive dependency is removed.

RESULT:

The EMPLOYEE relation was successfully decomposed into **3NF** by eliminating transitive dependencies.

Experiment 4: BCNF Decomposition**AIM:**

To verify whether a relation satisfies **Boyce-Codd Normal Form (BCNF)** and decompose it if necessary.

PROCEDURE:

1. List all functional dependencies.
2. Find the candidate keys.
3. Check each FD $X \rightarrow Y$.
4. Verify whether X is a superkey.
5. If X is not a superkey, decompose the relation.
6. Repeat until all relations satisfy BCNF.

Given:

$R(A, B, C, D)$

FDs:

$AB \rightarrow C$

$C \rightarrow D$

$D \rightarrow A$

Candidate keys:

AB, BC, BD

Check:

$AB \rightarrow C$

AB is a superkey $\rightarrow$ BCNF satisfied.

$C \rightarrow D$

C is not a superkey $\rightarrow$ BCNF violated.

Decompose:

$R_1(C, D)$

$R2(A,B,C)$

In R1:

$C \rightarrow D$

C is a key $\rightarrow$ BCNF.

In R2:

$AB \rightarrow C$

AB is a key $\rightarrow$ BCNF.

Final decomposition:

$R1(C,D)$

$R2(A,B,C)$

RESULT:

The relation was checked for BCNF and successfully decomposed where necessary.

Experiment 5: Lossless-Join Verification

AIM:

To verify whether a decomposition is **lossless** using the tabular/Chase algorithm.

PROCEDURE:

1. Write the original relation and its decomposition.
2. Create a tableau with one row for each decomposed relation.
3. Assign distinguished symbols to corresponding attributes.
4. Apply the given functional dependencies repeatedly.
5. Replace symbols when the FD conditions are satisfied.
6. Check whether a row containing all distinguished symbols is obtained.

Given:

$R(A,B,C)$

FD: $A \rightarrow B$

$R1(A,B)$

$R2(A,C)$

Final tableau:

Row	A	B	C
R1	aA	aB	b1
R2	aA	b2	aC

Apply:

$A \rightarrow B$

Both rows have the same value **aA** for A.

Therefore:

$b2 \rightarrow aB$

Final tableau:

Row	A	B	C
R1	aA	aB	b1
R2	aA	aB	aC

Since A determines B and A is the common attribute, the decomposition is **lossless**.

RESULT:

The decomposition was successfully tested using the Chase algorithm, and its lossless-join property was determined.

Experiment 6: 4NF Decomposition

AIM:

To identify multi-valued dependencies and decompose the TEACHER relation into **4NF**.

PROCEDURE:

1. Identify independent multivalued attributes.
2. Determine the multi-valued dependencies.
3. Check whether the determinant is a superkey.
4. If not, decompose the relation.
5. Create separate relations for the independent attributes.
6. Verify that the resulting relations satisfy 4NF.

Given:

TEACHER(TID, Subject, Hobby)

MVDs:

$TID \twoheadrightarrow Subject$

$TID \twoheadrightarrow Hobby$

Example:

T1 | DBMS | Music

T1 | DBMS | Cricket

T1 | OS | Music

T1 | OS | Cricket

Subject and Hobby are independent.

Decompose:

TEACHER_SUBJECT(TID, Subject)

TEACHER_HOBBY(TID, Hobby)

Result:

TEACHER_SUBJECT

T1 | DBMS

T1 | OS

TEACHER_HOBBY

T1 | Music

T1 | Cricket

The multivalued dependency is removed.

RESULT:

The TEACHER relation was successfully decomposed into **4NF** by removing multi-valued dependencies.

Experiment 7: 5NF Decomposition

AIM:

To identify join dependencies and decompose a relation into **Fifth Normal Form (5NF)**.

PROCEDURE:

1. Identify possible join dependencies.
2. Check whether the relation contains redundancy due to join dependencies.
3. Decompose the relation into smaller relations.
4. Join the decomposed relations to reconstruct the original relation.
5. Verify that no non-trivial join dependency remains.

Given:

SPJ(Supplier, Part, Project)

Assume there is a join dependency among Supplier, Part and Project.

Decompose into:

SP(Supplier, Part)

SJ(Supplier, Project)

PJ(Part, Project)

The original relation can be reconstructed using:

$SP \bowtie SJ \bowtie PJ$

The decomposition removes redundancy caused by the join dependency.

Therefore, the relations satisfy **5NF**.

RESULT:

The relation was successfully decomposed into **5NF**, reducing redundancy caused by join dependencies.

Experiment 8: Database Anomalies and 3NF

AIM:

To identify insertion, deletion, and update anomalies and normalize a college database into **3NF**.

PROCEDURE:

1. Examine the poorly designed relation.

2. Identify insertion anomalies.
3. Identify deletion anomalies.
4. Identify update anomalies.
5. Determine the functional dependencies.
6. Decompose the relation to remove transitive dependencies.
7. Verify the resulting relations are in 3NF.

Given:

COLLEGE(

StudentID,

StudentName,

CourseID,

CourseName,

InstructorID,

InstructorName

)

FDs:

$\text{StudentID} \rightarrow \text{StudentName}$

$\text{CourseID} \rightarrow \text{CourseName}, \text{InstructorID}$

$\text{InstructorID} \rightarrow \text{InstructorName}$

Transitive dependency:

$\text{CourseID} \rightarrow \text{InstructorID}$

$\text{InstructorID} \rightarrow \text{InstructorName}$

Therefore:

$\text{CourseID} \rightarrow \text{InstructorName}$

Decompose:

STUDENT(StudentID, StudentName)

COURSE(CourseID, CourseName, InstructorID)

INSTRUCTOR(InstructorID, InstructorName)

This removes:

- Insertion anomaly
- Deletion anomaly
- Update anomaly

RESULT:

All major database anomalies were identified and eliminated through **3NF normalization**.

Experiment 9: Attribute Closure

AIM:

To compute the closure of attribute sets using given functional dependencies.

PROCEDURE:

1. Start with the given attribute set.
2. Initialize its closure with those attributes.
3. Apply applicable functional dependencies.
4. Add newly determined attributes to the closure.
5. Repeat until no new attributes can be added.
6. Compare the closure with all attributes of the relation.

Given:

R(A,B,C,D,E)

$A \rightarrow B$

$B \rightarrow C$

$CD \rightarrow E$

$E \rightarrow A$

Calculate A^+ :

$A^+ = \{A\}$

$A \rightarrow B$

$A^+ = \{A,B\}$

$B \rightarrow C$

$A^+ = \{A,B,C\}$

Therefore:

$$A^+ = \{A, B, C\}$$

Calculate CD^+ :

$$CD^+ = \{C, D\}$$

$$CD \rightarrow E$$

$$CD^+ = \{C, D, E\}$$

$$E \rightarrow A$$

$$CD^+ = \{A, C, D, E\}$$

$$A \rightarrow B$$

$$CD^+ = \{A, B, C, D, E\}$$

Therefore:

$$CD^+ = \{A, B, C, D, E\}$$

RESULT:

The attribute closures were successfully computed using the given functional dependencies.

Experiment 10: Candidate Keys

AIM:

To determine all candidate keys of a relation using functional dependencies and attribute closure.

PROCEDURE:

1. List all attributes and functional dependencies.
2. Select possible attribute combinations.
3. Calculate the closure of each combination.
4. Check whether the closure contains all attributes.
5. Identify the combinations that determine the entire relation.
6. Remove unnecessary attributes to obtain minimal keys.

Given:

$$R(A, B, C, D, E)$$

$$A \rightarrow B$$

$$B \rightarrow C$$

$$CD \rightarrow E$$

$$E \rightarrow A$$

Calculate:

$$CD^+ = \{C,D\}$$

Using $CD \rightarrow E$:

$$CD^+ = \{C,D,E\}$$

Using $E \rightarrow A$:

$$CD^+ = \{A,C,D,E\}$$

Using $A \rightarrow B$:

$$CD^+ = \{A,B,C,D,E\}$$

Since CD^+ contains all attributes of R:

$$CD \rightarrow A,B,C,D,E$$

Therefore:

$$\text{Candidate Key} = \{C,D\}$$

RESULT:

All possible **candidate keys** of the given relation were successfully identified using attribute closure.